Grant Proposal Summary

Project Framing

We have built an end-to-end curation and extraction workflow for Alzheimer's disease genetics papers. The platform takes heterogeneous paper inputs, identifies relevant association tables, converts them into an ADVP-compatible structure, evaluates extracted records against reference ADVP data, and produces review-ready outputs for manual quality control.

Current demonstrated capabilities include:

- Multi-format ingestion from PMC table links, article pages, PDFs, Excel files, CSV files, TSV files, and HTML tables.
- Automatic discovery and ranking of PMC candidate tables using genetics- and association-oriented heuristics.
- Structured mapping of extracted table content into curated ADVP-compatible fields such as SNP, chromosome, base-pair position, allele fields, p-value, effect size, confidence interval, cohort, population, and sample-size metadata.
- Integrated evaluation outputs including row-match accuracy, field-level accuracy, mismatch reports, missing-field summaries, and fix-ready CSVs for downstream review.
- Browser-based editing and annotation so curators can correct extracted sheets while keeping links back to the paper and source table.

Example run reflected in the UI screenshots

PMID: 29274321

Paper source: https://pmc.ncbi.nlm.nih.gov/articles/PMC5938137/

Owner: Sherry

Tables selected for processing: 2

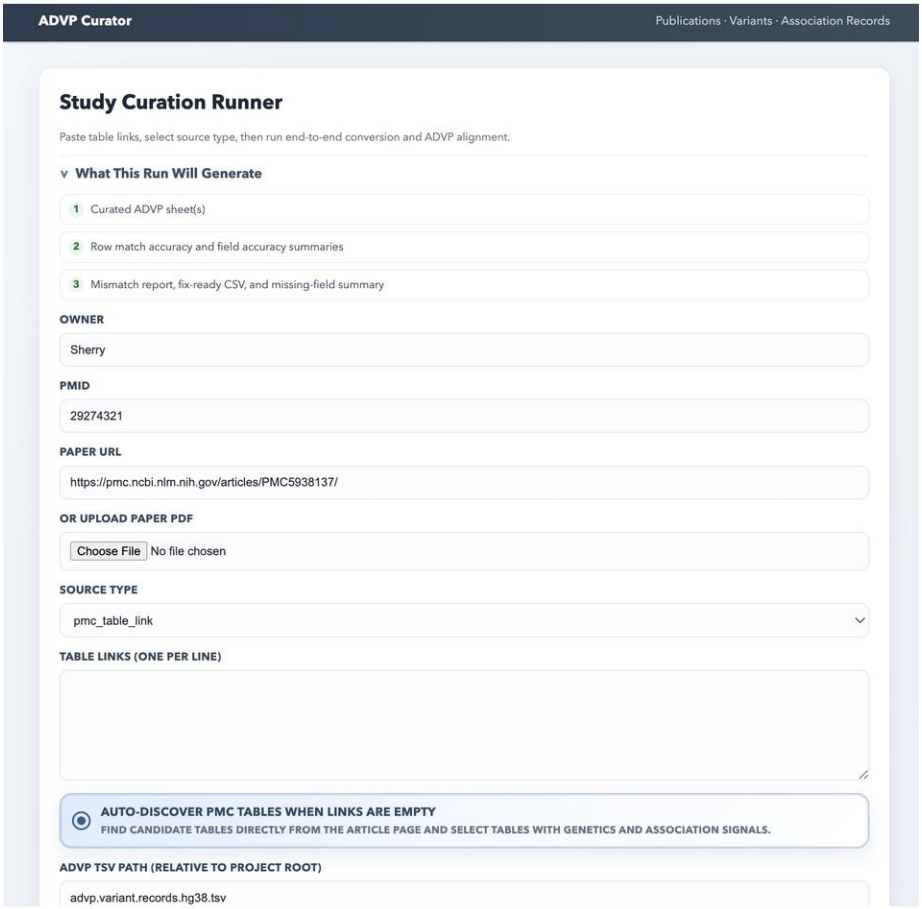
Curated spreadsheets generated: 2

Review artifact generated: pmid_29274321_fix_ready_20260408_232432.csv

The following figures document how the current website already embodies the major workflow components that can be described in the grant proposal as completed work.

Figure 1. Study Intake, Multi-Format Input, And PMC Table Discovery

Current Website Screenshot



What We Have Done So Far

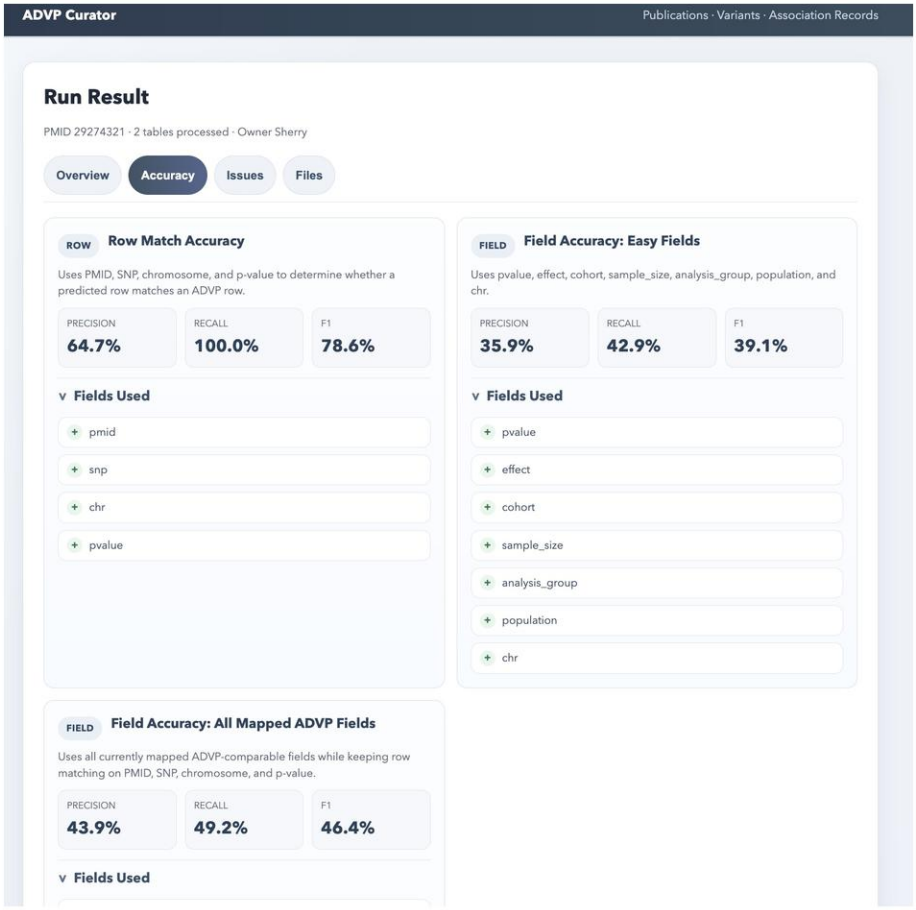
We built a web-based study curation runner that serves as the entry point for the full workflow. At this stage, a curator can provide a PMID, paper URL, uploaded PDF, source type, and table links, or allow the system to automatically discover candidate PMC tables when explicit links are not provided. This reflects substantial progress beyond a one-off extraction script because the system now supports standardized run setup, curator ownership, and reproducible execution across heterogeneous source formats.

How This Figure Supports the Proposal

- The interface operationalizes multi-format ingestion, which is a core completed contribution of the project.
- The PMC auto-discovery option demonstrates that we have already implemented logic to scan article content and identify candidate association tables without requiring full manual table selection.
- The generated-output checklist shows that each run already produces curated ADVP sheets plus downstream evaluation and review artifacts, not just raw extracted tables.

Figure 2. Row-Level And Field-Level Evaluation Against ADVP

Current Website Screenshot



What We Have Done So Far

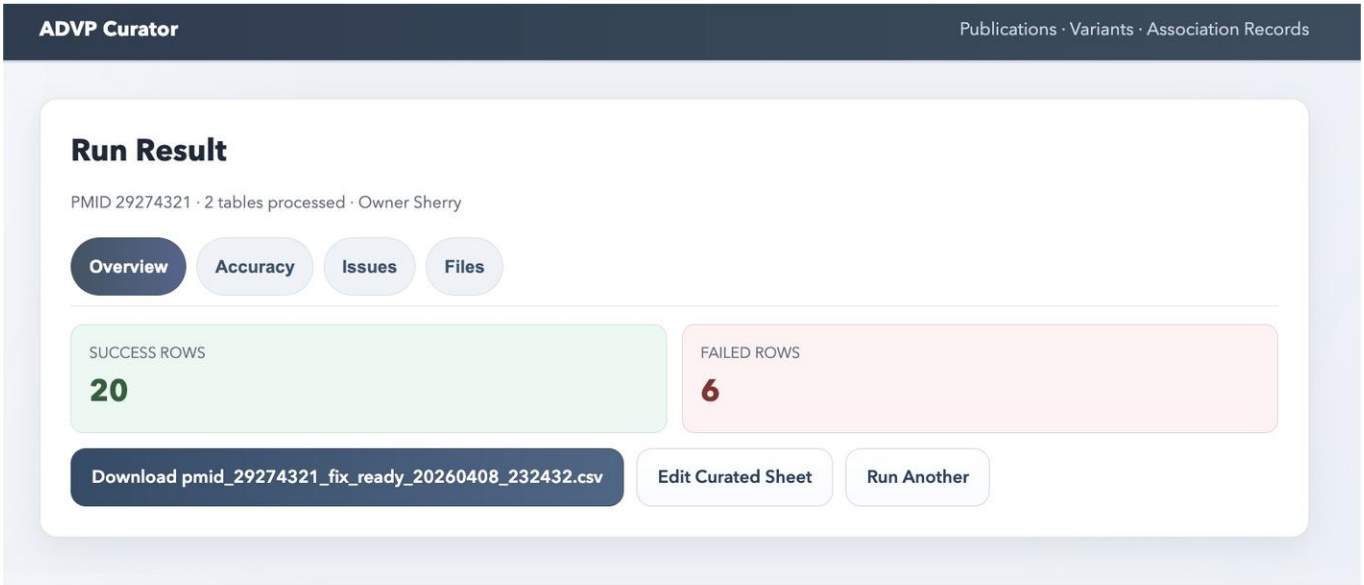
We built an evaluation layer that compares predicted structured outputs against ADVP reference records at both the row and field levels. The dashboard reports precision, recall, and F1 for row matching as well as field accuracy summaries for easier mapped fields and for the broader set of ADVP-comparable fields. This is important because it turns the system into a measurable curation platform, allowing us to quantify extraction quality, diagnose weaknesses, and prioritize refinement work.

How This Figure Supports the Proposal

- The row-match panel supports the claim that the workflow can benchmark whether extracted records correspond to expected ADVP association rows.
- The field-accuracy panels support the proposal's quality-control component by showing that harmonized attributes such as p-value, effect, cohort, sample size, analysis group, population, and chromosome are explicitly evaluated.
- The visible list of evaluated fields makes the system interpretable and reviewer-friendly, which is useful in a grant narrative focused on rigor and reproducibility.

Figure 3. Run-Level Review Outputs And Fix-Ready Deliverables

Current Website Screenshot



What We Have Done So Far

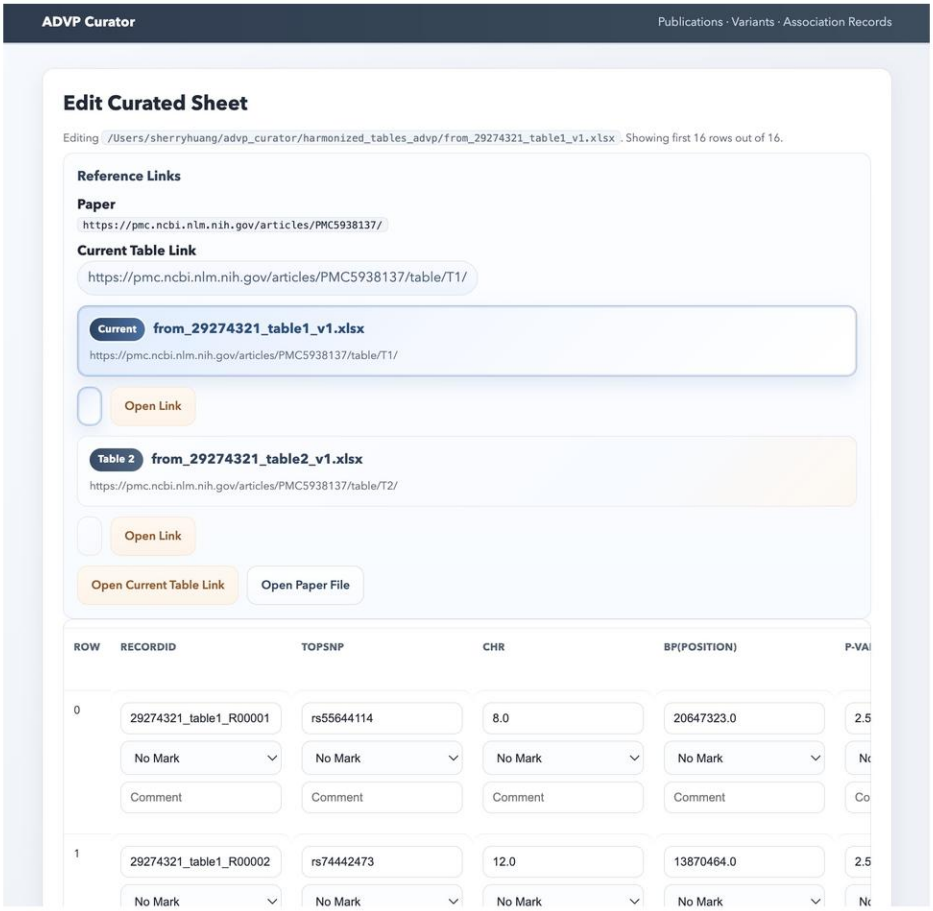
We built the workflow so that every completed run produces curator-facing review outputs rather than stopping at model predictions. The result page summarizes successful versus failed rows, exposes a downloadable fix-ready CSV, and provides direct transition points into additional review or correction steps. This completed functionality reduces manual curation effort because it packages discrepancies and incomplete outputs into artifacts that are immediately actionable.

How This Figure Supports the Proposal

- The success-versus-failed summary supports the claim that the platform provides run-level quality monitoring for practical curation work.
- The downloadable fix-ready CSV shows that we have already implemented downstream deliverables specifically designed for manual reconciliation and correction.
- The navigation from result summary to editing workflow demonstrates a closed-loop curation design rather than isolated extraction components.

Figure 4. Browser-Based Editing, Annotation, And Traceable Review

Current Website Screenshot



What We Have Done So Far

We also built a browser-based editing interface so curators can inspect extracted ADVP-compatible sheets, preserve links back to the paper and source table, and annotate or correct fields directly within the workflow. This is a key completed component because large-scale literature harmonization still requires human oversight, especially for ambiguous fields and heterogeneous table structures. By integrating editing into the same platform, we have created a practical human-in-the-loop environment for quality control and iterative refinement.

How This Figure Supports the Proposal

- The preserved paper and table links support traceability, allowing curators to verify each extracted record against its original evidence source.
- The row-level editing layout supports manual correction of harmonized values while keeping the ADVP-style structure intact.
- The mark-and-comment workflow supports annotation, error tracking, and future rule improvement, which strengthens the proposal's story around scalable semi-automated curation.